Reviewer Pack — Minimal Rank of Tropicalized Quaternion Algebras Bounds Comm…

Entry 9b4222dfe9d8 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Rank of Tropicalized Quaternion Algebras Bounds Communication Complexity of Symmetry Breaking

Entry ID: `9b4222dfe9d8` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-27 19:12:42 UTC

1. Conjecture

Field A (mathematical branch): Tropical Geometry (Quaternion Algebras) **Field B** (complexity object): Communication Complexity: Symmetry Breaking Communication Complexity

Statement:

> For every instance of the symmetry breaking communication complexity problem with n variables and t bits, there exists a tropicalized quaternion algebra representation such that the minimal rank of this algebra is upper bounded by a function f(t) = O(log(t)). Specifically, if S_B(n,t) denotes the symmetry breaking communication complexity for an instance with n variables and t bits, then for all instances, we have S_B(n,t) ≤ 2^(O(f(t))).

Rationale (proposer's reasoning):

> Quaternion algebras provide a rich structure that can encode complex symmetric operations. The tropicalization of these algebras may capture the essence of symmetry breaking communication complexity, potentially leading to exponential lower bounds.

Taxonomy category: `QuaternionAlgebraRepresentation` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `2b22db176930fd9c`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all tested instances with n variables and t bits (n ≤ 40, f(t) = O(log(t))), the minimal rank of the tropicalized quaternion algebra representation is at most O(f(t)) AND the symmetry breaking communication complexity S_B(n,t) is less than or equal to 2^(O(f(t))). The conjecture is falsified if either condition fails for any instance.

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM_1	LLM_2
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.95	HITS	UNCERTAIN
KARP_LIPTON	SAFE	1.00	UNCERTAIN	SAFE

4. Novelty audit

Verdict: `NOVEL` against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `"tropical geometry" AND "quaternion algebras" AND "communication complexity"` - `"symmetry breaking communication complexity" AND minimal rank" AND tropicalization"` - `"algebraic representation" AND "tropicalized quaternion algebras" AND O(log(t))`

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def gaussian_elimination(matrix):
        n = len(matrix)
        for i in range(n):
            max_row = i
            for j in range(i+1, n):
                if abs(matrix[j][i]) > abs(matrix[max_row][i]):
                    max_row = j
            matrix[i], matrix[max_row] = matrix[max_row], matrix[i]
            factor = matrix[i][i]
            for j in range(i, n + 1):
                matrix[i][j] /= factor
            for j in range(n):
                if i != j:
                    factor = matrix[j][i]
                    for k in range(i, n + 1):
                        matrix[j][k] -= factor * matrix[i][k]
        return matrix

    def min_rank(matrix):
        return len([row for row in gaussian_elimination(matrix) if any(row)])

    def symmetry_breaking_communication_complexity(n, t):
        # Placeholder function. Replace with actual implementation.
        return 2 ** (t // 4)
```

```
    n = random.randint(5, 40)
    t = random.randint(1, 2 * n)
    f_t = int(math.log(t))

    instance_matrix = [[random.choice([0, 1]) for _ in range(n)] for _ in range(n)]
    min_rank_value = min_rank(instance_matrix)
    comm_complexity = symmetry_breaking_communication_complexity(n, t)

    conjecture_holds = min_rank_value <= f_t and comm_complexity <= 2 ** (f_t)
    counterexample = "" if conjecture_holds else "mapping_undefined"

    return {
        "metric_name": "Symmetry Breaking Communication Complexity",
        "metric_value": comm_complexity,
        "instances_tested": 1,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    seeds = [int(arg) for arg in sys.argv[1:]] or [random.getrandbits(32) for _ in range(30)]

    results = []
    for seed in seeds:
        trial_result = run_trial(seed)
        print(f"TRIAL: {trial_result}")
        results.append(trial_result)

    mean_value = sum(result["metric_value"] for result in results) / len(results)
    std_value = math.sqrt(sum((result["metric_value"] - mean_value) ** 2 for result in results) /
    support_fraction = sum(1 for result in results if result["conjecture_holds"]) / len(results)

    if all(result["conjecture_holds"] for result in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_frac
    elif any(not result["conjecture_holds"] for result in results):
        first_failing_seed = next(seed for seed, result in zip(seeds, results) if not result["conj
        print(f"RESULT: FALSIFIED counterexample='mapping_undefined' first_failing_seed={first_fai
    else:
        print("RESULT: INCONCLUSIVE reason=unknown")
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94cdfc9a.py", line 70, in <module>
    trial_result = run_trial(seed)
                   ^^^^^^^^^^^^^^^

  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94cdfc9a.py", line 51, in run_trial
    min_rank_value = min_rank(instance_matrix)
                     ^^^^^^^^^^^^^^^^^^^^^^^^^
```

```
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94cdfc9a.py", line 40, in min_rank
    return len([row for row in gaussian_elimination(matrix) if any(row)])
                               ^^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_94cdfc9a.py", line 31, in gaussian_elim
    matrix[i][j] /= factor
    ~~~~~~~~~^^^
IndexError: list index out of range
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test crashed before producing data, which means we cannot verify whether the conjecture's conditions are met. | next: Re-run the test with a different seed or on a different system to ensure it completes successfully and provides results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	11162
2	preregistration	ollama_remote	glm4:latest	0	0	6244
3	novelty	ollama_remote	glm4:latest	0	0	4850
4	novelty	ollama_remote	glm4:latest	0	0	5266
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	17640
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	10178
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	7023
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8333
9	judge	ollama_remote	glm4:latest	0	0	8013

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 78708 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in `research/audit/9b4222dfe9d8.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/9b4222dfe9d8.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/9b4222dfe9d8.tar.gz` (if generated)